

Diagnostic Performance for Detection of Glaucomatous Structural Damage Using Pixelwise Analysis of Retinal Thickness Measurements

Hongli Yang,¹ Juan Reynaud,¹ Glen P. Sharpe,² Dawn Jennings,¹ Cindy Albert,³ Trinity Holthausen,³ Xiue Jiang,¹ Shaban Demirel,³ Steven L. Mansberger,³ Marcelo T. Nicolela,² Stuart K. Gardiner,³ Balwantray C. Chauhan,² Claude F. Burgoyne,¹ and Brad Fortune³

¹Optic Nerve Head Research Laboratory, Devers Eye Institute, Legacy Research Institute, Portland, Oregon, United States

²Department of Ophthalmology and Visual Sciences, Dalhousie University, Halifax, Nova Scotia, Canada

³Discoveries in Sight Research Laboratories, Devers Eye Institute, Legacy Research Institute, Portland, Oregon, United States

Correspondence: Hongli Yang, Devers Eye Institute, Legacy Research Institute, 1225 NE 2nd Ave, Portland, OR 97208-3950, USA; hyang@deverseye.org.

Received: March 17, 2024

Accepted: August 9, 2024

Published: October 9, 2024

Citation: Yang H, Reynaud J, Sharpe GP, et al. Diagnostic performance for detection of glaucomatous structural damage using pixelwise analysis of retinal thickness measurements. *Invest Ophthalmol Vis Sci.* 2024;65(12):17. <https://doi.org/10.1167/iov.65.12.17>

PURPOSE. To compare the diagnostic accuracy of thickness measurements of individual and combined macular retinal layers to discriminate 188 glaucomatous and 148 glaucoma suspect eyes from 362 healthy control (HC) eyes on a pixel-by-pixel basis.

METHODS. For this retrospective study, we manually corrected the segmentations of posterior pole optical coherence tomography (OCT) scans to determine the thickness of the nerve fiber layer (NFL), ganglion cell layer (GCL), inner plexiform layer (IPL), the ganglion cell complex (GCC), and the total neural retina (TR). For each eye, the total number of pixels with thickness values less than the fifth percentile of the HC distribution was used to create a receiver operating characteristic (ROC) curve for each layer and for layer combinations.

RESULTS. Using total abnormal pixel count criteria to discriminate glaucoma from HC eyes, the individual layers with the highest area under the ROC curve (AUC) were the NFL and GCL; IPL performance was significantly lower ($P < 0.05$). GCC had a significant higher AUC (94.3%) than individual the AUC of the NFL (92.3%) ($P = 0.0231$) but not higher than AUC of the GCL (93.4%) ($P = 0.3487$). The highest AUC (95.4%) and sensitivity (85.1%) at 95% specificity was found for the Boolean combination of NFL or GCL. The highest AUC is not significantly higher ($P = 0.0882$) than the AUC of the GCC but the highest sensitivity is significantly higher than the sensitivity of the GCC. This pattern was similar for discriminating between suspect and HC eyes ($P = 0.0356$).

CONCLUSIONS. Using pixel-based methods, the diagnostic accuracy of NFL and GCL exceeded that of IPL and TR. GCC had equivalent performance as NFL and GCL. The specific spatial locations within the posterior pole that exhibit best performance vary depending on which layer is being assessed. Recognizing this dependency highlights the importance of considering multiple layers independently, as they offer complementary information for effective and comprehensive diagnosis.

Keywords: glaucoma, macular retinal thickness, OCT macular thickness abnormality, nerve fiber layer, ganglion cell layer, pixel-based method

Glaucoma is the most common optic neuropathy worldwide and a major cause of irreversible blindness resulting from degeneration and death of retinal ganglion cells. Optical coherence tomography (OCT) enables objective three-dimensional visualization and morphometry of the anatomical changes caused by glaucoma, including loss of retinal ganglion cells (RGCs) and their axons, and the characteristic deformation of optic nerve head (ONH) tissues. The density of retinal ganglion cell bodies is normally highest within the macula, adjacent to the rim of the fovea, where OCT easily reveals this nuclear layer to be several times thicker than it is in more peripheral retina. This results in an increased dynamic range for detection and monitor-

ing of glaucoma by OCT layer thickness measurements. It may also facilitate enhanced diagnostic performance of these measurements by enabling image segmentation algorithms to be more accurate and reliable. Similarly, degeneration of retinal ganglion cell axon bundles is typically assessed by OCT and other imaging modalities in the peripapillary region, where the retinal nerve fiber layer (RNFL) is normally thickest and progressive thinning may be detected most reliably.^{1,2}

Another diagnostically important characteristic of retinal layer thinning caused by glaucoma is that it commonly occurs in stereotypical localized spatial patterns, such as the well-known "arcuate bundle defect." Confidence in detecting

glaucomatous structural loss is increased when all aspects correspond spatially, that is, thinning of corresponding sectors of the RGC layer, the inner plexiform layer (IPL), the RNFL and the ONH rim tissue, along with loss of associated microvasculature, which can be assessed noninvasively using OCT-angiography.³

Many studies have shown that glaucomatous structural damage can be detected with high accuracy using OCT measurements of tissue thinning in the macula area, either for individual inner retinal layers^{4–8} (such as the nerve fiber layer [NFL], ganglion cell layer [GCL], or IPL) or for combined layers, such as the ganglion cell-inner plexiform layer,^{6,9,10} or ganglion cell complex (GCC), defined as NFL, GCL, and IPL combined^{4,11–13}. However, most of these previous studies used either average thickness measurements for relatively wide spatial areas (global average over entire OCT scan area) or regional averages for sectors with predetermined shape/location. Similarly, few previous studies^{4,7} have compared diagnostic ability of individual retinal tissue layers (NFL, GCL, IPL, etc.). To our knowledge, no previous study has assessed and compared diagnostic performance on the basis of analyzing thickness measurements of individual retinal layers, combinations of layers, and using the data from individual pixels (rather than larger spatial regions). Some of this gap may be due to historical limitations of individual layer segmentation accuracy and the labor required to correct errors of automated segmentation algorithms, as well as previous limitations of OCT scan density. In this study, our goal was to assess the diagnostic ability of macular OCT scans to detect glaucomatous structural damage without using pre-determined spatial boundaries (sectors or cluster of points). We compared diagnostic performance of individual layers to each other and to combinations of layers in terms of discriminating eyes classified as having glaucoma (perimetric loss) from healthy eyes, and discriminating eyes classified as being glaucoma suspect (no perimetric loss) from healthy eyes.

METHODS

Study Participants

This cross-sectional study compares OCT, clinical and demographic data from participants enrolled in two separate longitudinal observational studies of glaucoma (GL) and glaucoma suspects (GLS), as well from a separate study of Healthy Controls (HC) that was designed to generate a normative reference database.^{14,15} For this study, OCT data from only one eye of each participant, selected at random, was used. The GL and GLS cohorts (see details below) were combined from two separate ongoing studies: (1) the Portland Progression Project,^{16–18} a longitudinal study, supported by the National Institutes of Health, based at the Legacy Devers Eye Institute in Portland, Oregon, whose aims are to determine the risk factors predicting the rate of progression and to evaluate and improve relevant diagnostic tools in ocular hypertensive and early glaucoma eyes, and (2) longitudinal glaucoma studies at the Eye Care Centre, Nova Scotia Health Authority, Halifax, Nova Scotia, Canada. The aims of Halifax longitudinal study, supported by the Canadian Institutes of Health Research, are to longitudinally assess the rate of progression in glaucoma eyes using OCT structural parameters and visual field-testing parameters. Enrollment of participants for both the Portland and Halifax studies was based not only on these criteria but also on the willingness and likelihood of each recruit to remain in

the longitudinal study, itself a subjective criterion; however, this subjectivity is not based on any ocular findings. Data for the HC cohort was obtained from a previous multi-centered normative study^{14,15} (eight centers total: five in the United States, two in Germany, and one in Canada). Subjects for HC cohort were recruited by advertisement in local media, on bulletin boards, and from a registry of subjects who had taken part in other studies. Approximately equal numbers of subjects were recruited in each decade group from 20 to 90 years.

All procedures in all of these studies adhered to the tenets of the Declaration of Helsinki, complied with the Health Insurance Portability and Accountability Act of 1996, and were approved and monitored by the Institutional Review Board or its equivalent at each center. All participants provided written informed consent after having the risks and benefits of participation explained to them.

Inclusion criteria for the HC subjects included age 18 to 90 years, no history of glaucoma, intraocular pressure (IOP) ≤ 21 mm Hg, best-corrected visual acuity $\geq 20/40$, refraction less than ± 6.00 diopters sphere and ± 2.00 diopters cylinder; Glaucoma Hemifield Test within normal limits (from standard automated perimetry, using the HFA II perimeter [Carl Zeiss Meditec Inc, Dublin, CA, USA], 24-2 test pattern and SITA Standard test strategy), and the availability of acceptable ONH, circumpapillary RNFL, and macular OCT scans. Exclusion criteria were unusable stereo photographs or insufficient OCT scan quality (scan quality score < 20), clinically abnormal optic disc appearance, any intraocular surgery (except uncomplicated cataract surgery) or any vitreal, retinal, choroidal, or neuro-ophthalmological disease.^{14,15}

Our reference standard test for glaucoma included the following functional test as described below.¹⁹ GL eyes (combined from Portland Progression Project and Halifax cohorts) had a clinical diagnosis of glaucoma as judged by the patient's clinician and confirmed for this study by visual field criteria of pattern standard deviation > 1.99 dB) or GHT outside normal limits on two consecutive tests with the Humphrey Field Analyzer 24-2 pattern. GLS eyes (derived from the Portland Progression Project only) had ocular hypertension (untreated IOP greater than 22 mm Hg), or a suspicious optic disc appearance (neuroretinal rim notching or narrowing; or disc hemorrhage) along with one or more other risk factors for glaucoma (e.g., age > 70 , systemic hypertension, migraine, diet-controlled diabetes, peripheral vasospasm, African ancestry, a self-reported family history of glaucoma, or glaucoma in the fellow eye), but no glaucomatous loss on repeated visual field testing, as described above for the HC cohort. Exclusion criteria for GL/GLS eyes were other causes of significant visual field loss, or inability to produce reliable test results.^{16,18,20} For subanalyses reported in the Results section, GL eyes were further subdivided into two sub-groups: early GL (MD ≥ -4 dB) and moderate or advanced GL (MD < -4 dB).

OCT Acquisition, Segmentation, and Pixel-Based Quantification of Macular Layer Thickness

In this study, the index test¹⁹ is defined as pixel-wise quantification of different macular layer thickness values, adjusted for age and distance between the fovea and ONH, as detailed in this section. The OCT acquisition protocol was identical for all GL, GLS, and HC eyes across all three studies and has been described in detail previously.^{17,21} Briefly, spectral domain OCT imaging of the ONH, circumpapillary

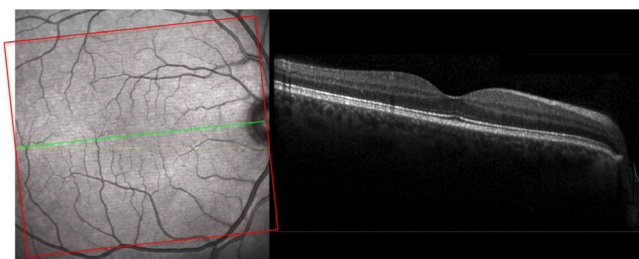


FIGURE 1. FoBMO macular scan pattern and the macular B-scan (*Left*) 61 B-scans, 768 A-lines / B-scan, $30^\circ \times 25^\circ$ area centered on the fovea. (*Right*) The B-scan corresponds to scan on the left (the scan location highlighted with green line).

RNFL and macula was performed on each study eye using the same type of instrument and software version (Spectralis; Heidelberg Engineering GmbH, Heidelberg, Germany; HEYEX software version 5.4c, Glaucoma Module Premium Edition). For each eye, before image acquisition, refractive correction and values of anterior corneal surface curvature (measured by keratometry) were entered into the OCT instrument database. The OCT scan acquisition protocol used the instrument's interactive Anatomic Positioning System whereby the operator identifies the position of the fovea using live B-scan views, and an estimate of the center of the ONH by marking a pair of BMO points within each of two orthogonal B-scans crossing the ONH. These steps thus also established the fovea-BMO centroid (FoBMO) axis for each eye, which determines the orientation of three subsequent scan types obtained and used for this study.¹⁷ (Glaucoma Module Premium Edition In: Spectralis OCT Glaucoma brochure: Heidelberg Engineering).

The first scan type collected was the ONH scan using an enhanced depth imaging mode, which is comprised of 24 radially oriented B-scans, each containing 768 A-lines, spanning 15° total width and centered on the ONH BMO centroid. The real-time averaging was set to 25 repetitions.

A separate circular B-scan (12° diameter, 1536 A-lines, average of 100 repetitions), which was also centered on the ONH BMO centroid using the anatomic positioning system, was used to measure circumpapillary RNFL thickness. Finally, the third OCT scan type acquired was the "posterior pole horizontal" pattern, which is the primary focus of this study. This pattern covers a rectangular area of the posterior pole, 30° wide by 25° height, centered on the fovea (identified in the anatomic positioning system) and oriented parallel to the FoBMO axis according to the anatomic positioning system protocol (Fig. 1). It is comprised of 61 equally spaced B-scans ($\sim 120 \mu\text{m}$ apart) each containing 768 A-lines, spanning 30° total width and averaged in real time using 10 repetitions.

All OCT scans were required to have an averaged scan quality score ≥ 20 and be free of large truncation or mirror artifacts. All individual B-scans within each of the posterior pole horizontal scans were examined to meet these inclusion criteria. Approximately 5% of OCT scans were excluded for failing to because of one or more of these criteria. For all OCT scan types, the image segmentation began with the instrument's automated set, which was then inspected and manually corrected as necessary by trained observers using custom software (ATL 3D Suite, Portland, OR, USA).^{17,22,23} For the posterior pole horizontal scans covering the macula, five retinal layer boundaries were used to calculate layer thicknesses along each A-line, including the internal limiting membrane (ILM), external boundaries of the NFL, GCL, inner plexiform layer (IPL), and neurosensory retina [i.e., the anterior aspect of the retinal pigment epithelium (RPE)] (Fig. 2). Thus a continuous thickness map (containing a maximum of 768×61 -pixel values) was generated for each of three individual layers of the macular inner retina: NFL, GCL, and IPL, as well as for the ganglion cell complex (GCC, from ILM to posterior boundary of IPL), and total neural retinal thickness (TR, from ILM to the inner boundary of RPE). For each eye, a circular mask with 3° radius centered on the BMO centroid was used to exclude A-lines within or immediately adjacent to the ONH from all calculations of retinal layer

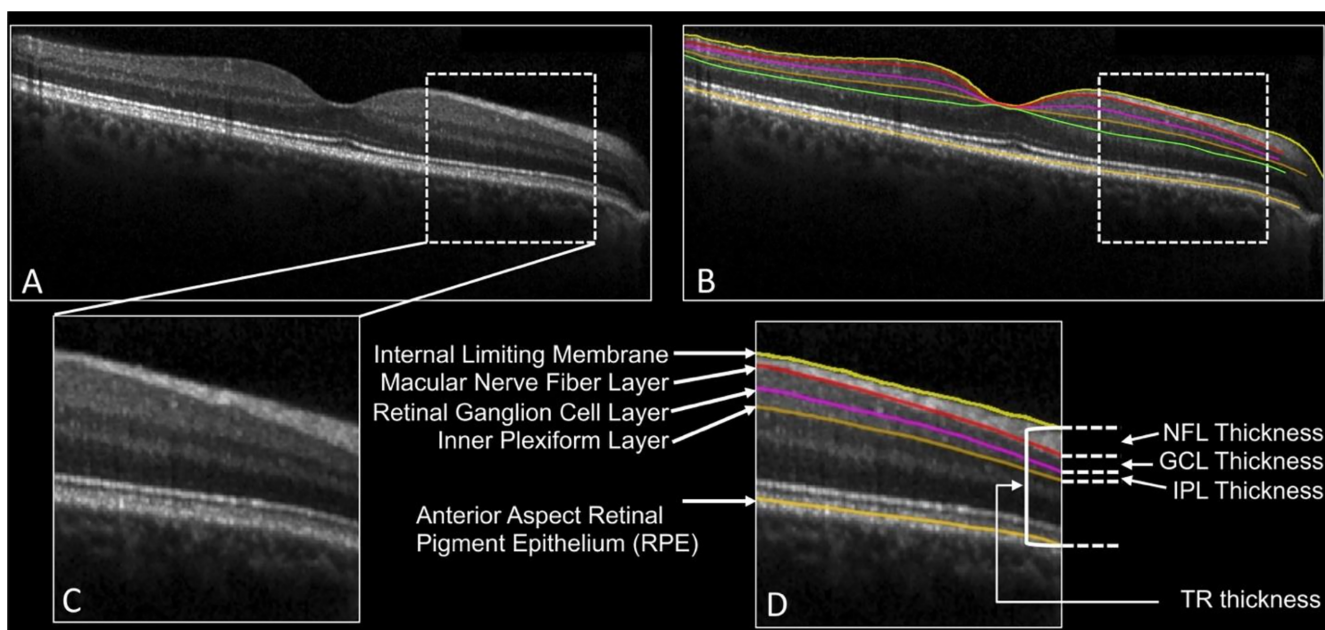


FIGURE 2. Macular B-scan Segmentation. Segmentations within each B-scan were manually corrected and thickness values were measured for individual inner retinal layers at each A-scan location: NFL, GCL, IPL, GCC, and the TR.

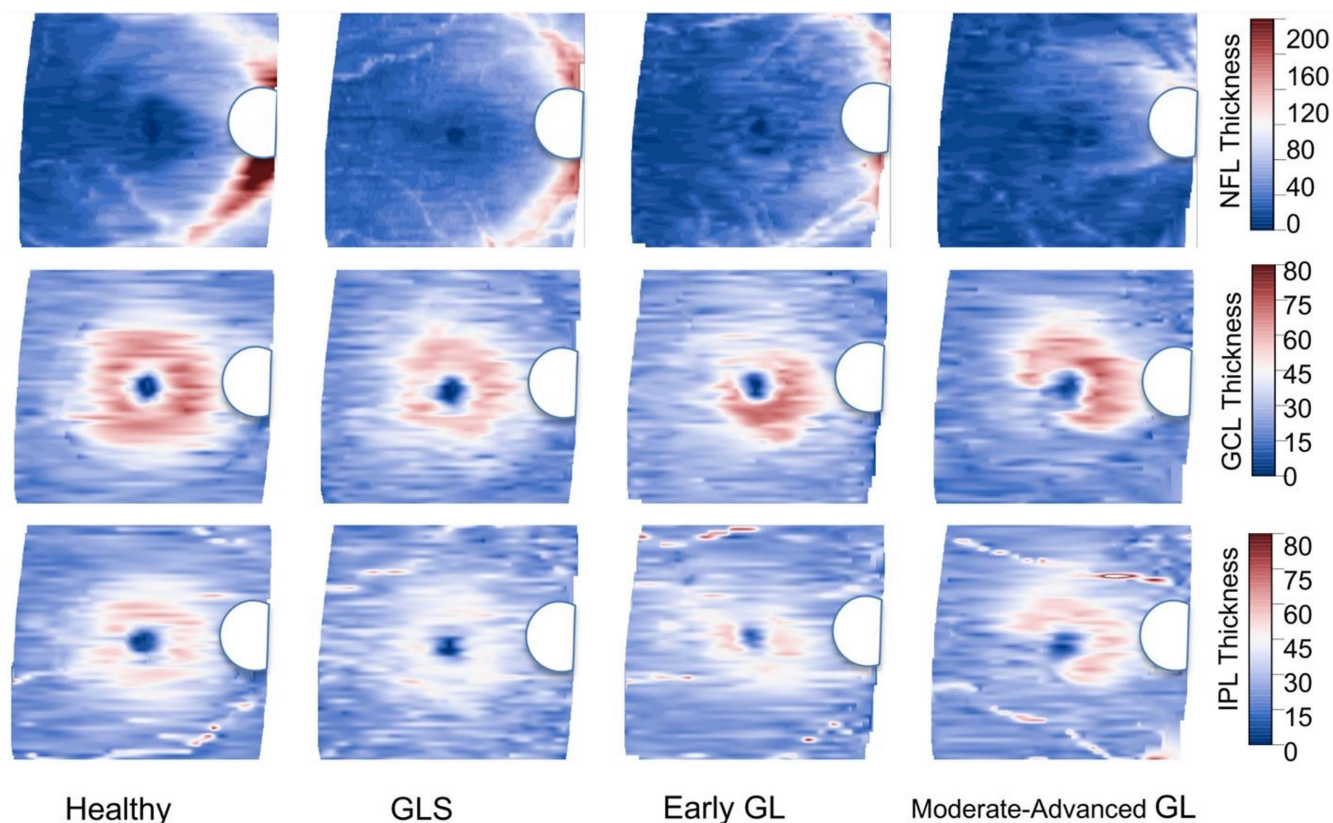


FIGURE 3. Individual layer thickness of a representative healthy, glaucoma suspect, early glaucoma, moderate-advanced glaucoma eye. Data was color-coded and printed on a continuous map with interpolated data from 768×61 pixels for each individual eye. From top row to bottom row: NFL thickness, GCL thickness, and IPL thickness.

thickness values. A visualization of the ONH mask can be seen in Figure 3.

Data Analysis

Because the primary purpose of this study was to compare the diagnostic performance of various individual inner retinal layers, as well as combinations of layers, we applied a receiver operating characteristic (ROC) curve analysis and determined the area under the ROC curve (AUC), as well as the sensitivity at 95% specificity to compare the accuracy of each layer or layer combination to discriminate GL and GLS eyes from HC eyes. We also used a variety of criteria to define a thickness “abnormality,” each of which was then used to generate separate ROC curves (see below) to assess the potential advantages of each criterion.

The first step of the analysis was to adjust each observation (each thickness value at each pixel) by age and distance between the ONH and fovea (i.e., the linear FoBMO distance), given that both factors are well known to influence peripapillary²⁴ and macular retinal thickness²¹ (our data showed age and FoBMO distance were significantly associated with macular layer thickness in more than 50% of pixels for most of the individual and combination layers). Thus each measured thickness value at each pixel was adjusted to the equivalent value for age 70 years and the average FoBMO (Fovea-BMO centroid) distance of the HC eyes, based on linear regression among the HC eyes. For this cohort of HC eyes, the slope of thickness versus age for the NFL, GCL, IPL, GCC and TR, respectively, was -0.02 , -0.06 , -0.04 , -0.12 ,

and $-0.24 \mu\text{m}/\text{year}$. Using the adjusted thickness values for each pixel, we defined an “OCT structural abnormality” to be any observation that was below the fifth percentile of the HC eye distribution. We then counted the number of abnormal pixels within each eye and generated ROC curves on the basis of those results, as well as other criteria, as follows:

- (1) Total abnormal pixel count of each individual eye/layer. This approach represents primarily the lateral spatial extent of an abnormality, but would not necessarily capture increasing depth of anatomical defects as severity worsens. We generated a family of ROC curves based on the abnormal pixel count, whereby the performance of each parameter (individual inner retinal layer thickness) was represented by one ROC curve.
- (2) Total abnormal pixel count of each individual eye using combinations of layers. This approach used both physical layer combinations, GCC, and TR, as well as the results of Boolean combinations of individual inner retinal layers, in order to count the number of abnormal pixels per eye, including: (2A) “NFL and GCL” (if both layer thicknesses were abnormally thin at a given pixel, this parameter was defined as being abnormal at that pixel); (2B) “GCL and IPL”; (2C) “NFL or GCL” (if NFL or GCL thickness was abnormally thin at a given pixel, then this parameter was defined as being abnormal for that pixel); (2D) any two of the three layers: NFL, GCL, IPL.

Using this criterion of Count of all abnormal pixels (i.e., those below 5th percentile of normal), we generated a ROC curve for each of the individual macular inner retinal layers (NFL, GCL, and IPL), as well as for the parameters of physically combined layers (GCC and TR) and each of the Boolean logical combinations described above for criterion number 2. We estimated sensitivity at 95% specificity and area under the ROC curve as well as their 95 % confidence intervals.

- (3) To potentially capture highly localized defects, which may be more severe (greater depth) but of lesser lateral extent, we used criteria based on the cumulative Z-score of all abnormal pixels within each eye. Thus, if two different eyes each had a defect with the same lateral extent, the one with the deeper defect (thinner tissue) would have a worse cumulative Z-score. The Z-score for each parameter (layer thickness) at each pixel was calculated assuming a standard normal distribution, for example, using NFL thickness at a given pixel:

$$Z\text{-score} = \frac{NFLT(obs) - NFLT \text{ mean of HC}}{\text{Standard Deviation of HC}}$$

The cumulative Z-score was defined as the sum of these pixel-wise Z-scores for all abnormal pixels (i.e., those below fifth percentile of normal). As before, we generated a ROC curve for each of the individual macular inner retinal layers (NFL, GCL, and IPL), as well as for the parameters of physically combined layers (GCC and TR) and each of the Boolean logical combinations described above for criterion number 2. We determined sensitivity at 95% specificity and area under the ROC curve, then also estimated their 95% confidence intervals using bootstrapping.

All statistical analyses were performed in R version 3.1.3 (The R Foundation for Statistical Computing, Vienna, Austria). For both AUC and sensitivity at 95% specificity, 95% confidence intervals were empirically generated using 2000 bootstrap resamplings of the data. The AUC of the NFL was set as the reference and compared with the AUC of all other

criteria using the DeLong method.²⁵ Similarly, the sensitivity at 95% specificity for the NFL was set as the reference and compared with the sensitivity of all other criteria using the bootstrap method.²⁶

RESULTS

A total of 698 eyes of 698 participants were included in this study, comprising 362 HC eyes (Age range from 20 to 90), 148 GLS eyes (Age range from 38 to 92) and 188 GL eyes (Age range from 49 to 92) (Table 1). The distribution of visual field Mean Deviation (VF MD) among the GL and GLS eyes is depicted in Supplemental Figure S1. Among the GL eyes, 86 (45.7%) demonstrated early VF defects (VF MD $\geq$ -4 dB, median -1.92 dB, range -3.94 to 0.72 dB), and 102 eyes (54.3%) demonstrated moderate-advanced VF defects (VF MD < -4 dB, median -7.27 dB, range -27.01 to -4.02 dB). Compared to the HC group, GL subjects were older and GL eyes had a longer FoBMO distance, as well as thinner global average values for macular NFL, GCL, IPL, GCC and TR thickness (Table 1). The GLS group was also older than the HC group, had a more negative spherical equivalent refractive error, longer FoBMO distance, slightly larger global average thickness of NFL, but smaller global average values for macular GCL, and TR thickness (Table 1). Compared to the GLS group, GL subjects were older and GL eyes had lower IOP, shorter FoBMO distance, and thinner global average values for macular NFL, GCL, IPL, GCC and TR thickness. Compared to HC eyes, GL and GLS eyes had very were very slightly, but significantly lower Scan Quality Scores. However, further evaluation indicated that Scan Quality Score was not a confounding factor for individual layer thickness: in a multivariate regression model, individual layer thickness values were associated with age only but not with Scan Quality Score. The demographics of early GL subjects were very similar to moderate-advanced GL subjects, however, early GL eyes had thicker global average values of macular NFL, GCL, IPL, GCC, and TR thickness, on average (Table 2).

TABLE 1. Demographic, Ocular and Visual Field Characteristics of the Study Participants

	HC Eyes (n = 362)	GLS Eyes (n = 148)	GL Eyes (n = 188)	P Value ¹
Age (y), (SD)	50.62 (17.5)	68.6*** (10.1)	70.68*** (9.65)	<0.0001
Female gender	202 (55.8%)	81 (54.7%)	95 (50.5%)	0.4944
Left eye	181 (50.0%)	66 (44.6%)	91 (48.4%)	0.5409
IOP (mm Hg), (SD)	14.53 (2.7)	17.57*** (3.69)	13.98*** (4.41)	<0.0001
Spherical equivalent (D), (SD)	-0.47 (1.82)	-0.84* (2.07)	-0.75 (1.9)	0.0808
Visual field MD (dB), median (range)	N/A	0.34 (-4.50 to 2.93)	-4.37*** (-27.01 to 0.72)	<0.0001
Visual field PSD (dB), median (range)	N/A	1.61 (1.06 to 4.90)	5.73*** (1.66 to 15.94)	<0.0001
BMO area (mm ²), (SD)	1.83 (0.38)	1.81 (0.41)	1.90 (0.44)	0.1108
FoBMO distance (mm), (SD)	4.39 (0.29)	4.29** (0.26)	4.28*** (0.28)	<0.0001
Global NFL (μm), (SD)	46.91 (4.44)	48.51* (7.68)	35.24***## (7.47)	<0.0001
Global GCL (μm), (SD)	28.82 (2.43)	26.51*** (2.75)	23.87***## (2.93)	<0.0001
Global IPL (μm), (SD)	25.44 (2.09)	25.42 (2.57)	23.52***## (2.34)	<0.0001
Global GCC (μm), (SD)	101.19 (7.43)	100.50 (11.09)	82.53***## (11.14)	<0.0001
Global TR (μm), (SD)	274.54 (12.64)	268.22*** (14.36)	253.85***## (15.86)	<0.0001
OCT Quality Score (dB), (SD)	27.7 (3.2)	26.2*** (3.2)	25.7*** (2.6)	<0.0001

ANOVA, analysis of variance; MD, mean deviation; N/A (not applicable); PSD, pattern standard deviation.

Data are mean (SD) values except for the number of subjects, gender, and eye (depicted as discrete values) and visual field indexes (depicted as median and range).

¹ Differences for continuous variables were assessed by ANOVA for means and by median test for median values. Differences for dichotomous variables were tested with a χ^2 test. Bold: $P < 0.05$. Pairwise comparisons between GL and HC eyes and between GLS and HC eyes were performed by linear regression (* $P < 0.05$; ** $P < 0.001$; *** $P < 0.0001$). Pairwise comparisons between GL and GLS eyes were also performed by linear regression (* $P < 0.05$; ** $P < 0.001$; *** $P < 0.0001$).

TABLE 2. Demographic and Ocular Characteristics of Early and Moderate–Advanced GL Subjects and Eyes

	GL Eyes		P Value ¹
	Early (n = 86)	Moderate–Advanced (n = 102)	
Age (y), (SD)	69.52 (8.96)	71.66 (10.1)	0.131
Female gender	43 (50.0%)	52 (51.0%)	1.000
Left eye	35 (39.8%)	56 (54.9%)	0.072
IOP (mm Hg), (SD)	15.44 (3.96)	12.75 (4.41)	<0.0001
Spherical equivalent (D), (SD)	−0.77 (2.03)	−0.73 (1.79)	0.8875
Visual field MD (dB), median (range)	−1.92 (−3.94 to 0.72)	−7.27 (−27.01 to −4.02)	<0.0001
Visual field PSD (dB), median (range)	3.28 (1.66 to 9.54)	8.75 (2.62 to 15.94)	<0.0001
BMO area (mm ²), (SD)	1.91 (0.44)	1.89 (0.45)	0.792
FoBMO Distance (mm), (SD)	4.26 (2.47)	4.29 (2.98)	0.432
Global NFL (μm), (SD)	38.38 (7.67)	32.59 (6.18)	<0.0001
Global GCL (μm), (SD)	25.10 (2.43)	22.82 (2.93)	<0.0001
Global IPL (μm), (SD)	24.02 (2.30)	23.10 (2.30)	0.0066
Global GCC (μm), (SD)	87.51 (10.70)	78.34 (9.71)	<0.0001
Global TR (μm), (SD)	260.82 (14.46)	247.97 (14.61)	<0.0001
OCT Quality Score (dB), (SD)	25.7 (3.6)	25.8 (2.7)	0.86

ANOVA, analysis of variance; MD, mean deviation; N/A (not applicable); PSD, pattern standard deviation.
Data are mean (SD) values except for the number of subjects, gender, and eye, (depicted as discrete values) and visual field indexes, (depicted as median and range).
¹ Differences for continuous variables were assessed by ANOVA for means and by median test for median values. Differences for dichotomous variables were tested with a Chi-square test. Bold: *P* < 0.05.

Discriminating Glaucoma (GL) From Healthy Control (HC) Eyes

Representative examples of one eye each from the HC, GLS, early GL and moderate-advanced GL groups

are shown in Figure 3 for three of the individual retinal layer thickness maps, and in Figure 4 for their corresponding z-score (“abnormality”) maps. These examples demonstrate the tendencies for asymmetry of damage across the horizontal midline as well as differ-

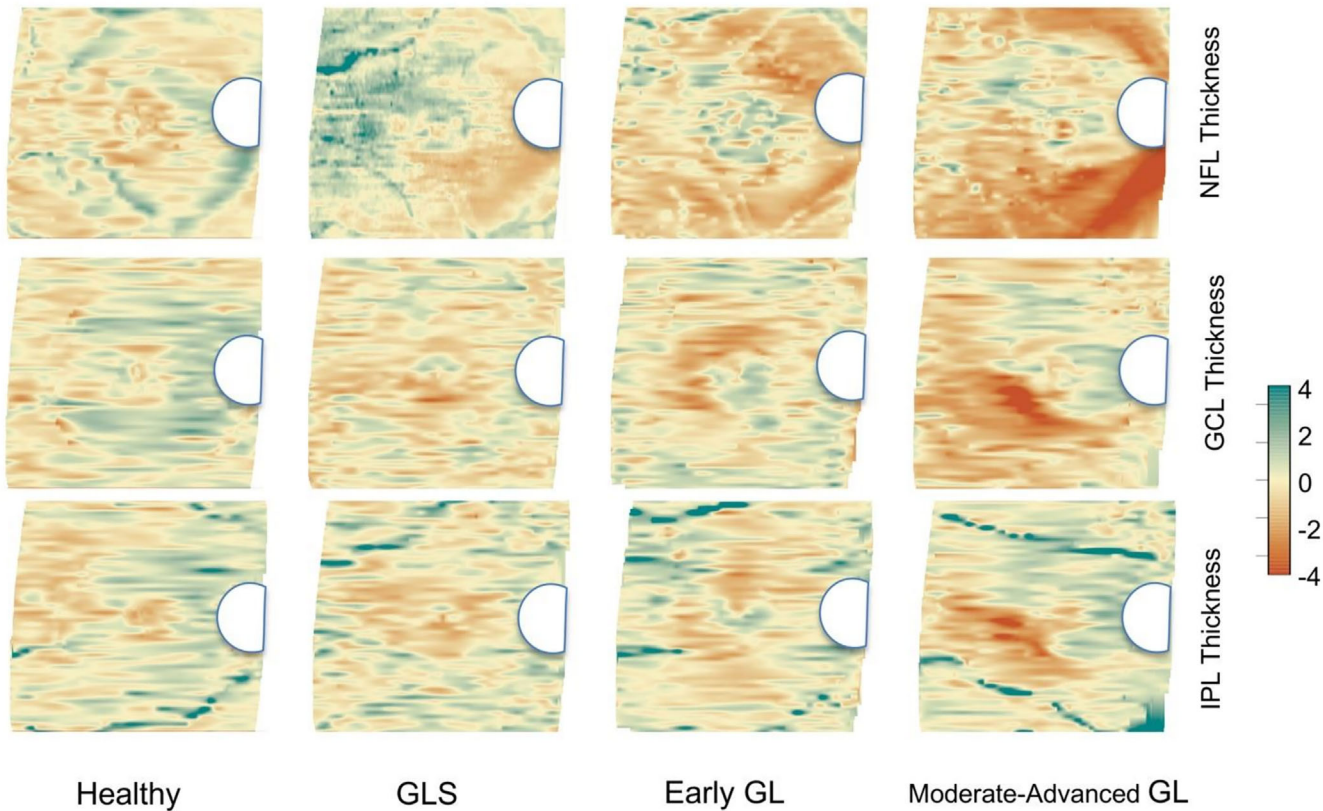


FIGURE 4. Z-score of the Individual layer thickness of a representative healthy, glaucoma suspect, early glaucoma, moderate-advanced glaucoma eye. Data was color-coded and printed on a continuous map with interpolated data from 768 × 61 pixels for each individual eye. From top row to bottom row: NFL thickness, GCL thickness, and IPL thickness.

TABLE 3. Single and Combined Layer Thickness Using Abnormal Pixel Count Detection Criteria in the GL and GLS Eyes

	Glaucoma		Glaucoma Suspect	
	Sensitivity at 95% Specificity	AUC	Sensitivity at 95% Specificity	AUC
Global Single Layer				
NFL	79.5 (69.7–86.7)	92.3 (89.3–95.3)	34.5 (26.4–43.2)	58.9 (52.3–65.4)
GCL	74.5 (65.4–83) (<i>P</i> = 0.3996)	93.4 (91.2–95.6) (<i>P</i> = 0.4887)	35.1 (23.6–48) (<i>P</i> = 0.8641)	75.8 (70.6–80.9) (<i>P</i> = 0.0009)
IPL	58.5 (43.1–66.5) (<i>P</i> = 0.0037)	88.7 (85.7–91.6) (<i>P</i> = 0.0303)	18.9 (8.8–26.4) (<i>P</i> = 0.0093)	65.2 (59.8–70.6) (<i>P</i> = 0.2341)
Combined Layer				
GCC (all NFL, GCL, IPL)	77.7 (68.6–87.2) (<i>P</i> = 0.7463)	94.3 (92.2–96.4) (<i>P</i> = 0.0232)	14.9 (8.1–25) (<i>P</i> = 0.0006)	58.2 (52.5–64) (<i>P</i> = 0.9172)
GCL and IPL	80.3 (72.3–86.7) (<i>P</i> = 0.9239)	93.1 (90.4–95.8) (<i>P</i> = 0.6196)	29.1 (18.9–37.8) (<i>P</i> = 0.4102)	63.8 (57.7–69.9) (<i>P</i> = 0.3827)
NFL and GCL	81.9 (73.9–87.8) (<i>P</i> = 0.5683)	93.4 (90.9–95.8) (<i>P</i> = 0.3549)	18.2 (12.2–29.1) (<i>P</i> = 0.0063)	60.1 (54.8–65.4) (<i>P</i> = 0.8285)
Any 2 of 3 (NFL, GCL, IPL)	83 (75–88.8) (<i>P</i> = 0.3585)	94.5 (92.4–96.7) (<i>P</i> = 0.1003)	27.7 (17.6–37.2) (<i>P</i> = 0.3630)	62.9 (56.8–69) (<i>P</i> = 0.4940)
NFL or GCL	85.1 (78.7–90.4) (<i>P</i> = 0.1947)	95.4 (93.6–97.2) (<i>P</i> = 0.0066)	29.1 (18.9–40.5) (<i>P</i> = 0.5628)	68.7 (63–74.4) (<i>P</i> = 0.0803)
TR	55.9 (44.1–67.6) (<i>P</i> = 0.0017)	90.3 (87.7–92.9) (<i>P</i> = 0.1758)	15.5 (8.1–25.7) (<i>P</i> = 0.0030)	70.0 (65.1–74.9) (<i>P</i> = 0.0307)

AUC (%) and sensitivity (%) at 95% specificity as well as 95% confidence interval are listed in the table. Every criterion is compared against NFL, and *P* value is listed.

ent spatial patterns of damage for different inner retinal layers.

The results for diagnostic performance based on total abnormal pixel count (Criteria 1 and 2) are listed in Table 3 and plotted as ROC curves in Figure 5. For GL eyes, the individual inner retinal layers with the best performance were the NFL and GCL; the GCL had the highest AUC (93.4% [91.2%–95.6%]), and second-highest sensitivity (at 95% specificity, 74.5% [65.4%–83.0%]). The NFL had a higher sensitivity (79.5% [69.7%–86.7%]) but a slighter lower AUC (92.3% [89.3%–95.3%]), although neither of the values for NFL was significantly different than those for the GCL (sensitivity *P* = 0.3996; AUC: *P* = 0.4887). Diagnostic performance of the IPL in GL eyes was significantly lower than that of either NFL or GCL based on values for both AUC (88.7% [85.7%–91.6%], vs. NFL *P* = 0.0303; vs. GCL *P* < 0.0001) and sensitivity (58.5% [43.1%–66.5%], vs. NFL *P* = 0.0037; vs. GCL *P* = 0.0397).

Among the combined layer parameters, all of those derived from the total abnormal pixel count of inner retinal layers (Criterion 2) had similar performance to GCL and NFL by AUC (Table 3). The highest AUC overall for GL eyes was found for the Boolean combination criterion of NFL or GCL (95.4%), which was significantly better than the AUC of either individual layers NFL (*P* = 0.0066) or GCL (*P* = 0.0019) but not significantly higher than that for GCC (*P* = 0.0882); the sensitivity of the Boolean NFL or GCL (85.1%) was also highest overall for GL eyes, and this was significantly better than GCC (77.7%) (*P* = 0.0356). It should be noted that an additional analysis of partial AUC, using only the portion of the ROC with specificity >0.80 and >0.90, produced the same pattern of results, which is evident by visual inspection of that portion of the ROC curves in Figures 5 and 6.

The outcomes based on analysis of Cumulative Z-scores (Criterion 3) are listed in Supplemental Table S1. The pattern of results of the cumulative Z-score analysis of GL eyes is similar to that found using the total abnormal pixel count. Yet interestingly, although none of the differences between

these two approaches were statistically significant, there was a strong trend toward the performance of the Cumulative Z-score approach being slightly superior to the total pixel count approach. For example, using the Cumulative Z-score of all abnormal pixels (i.e., Criterion 3: all pixels with parameter value below fifth percentile of HC eyes), the best-performing individual layers (NFL and GCL) and the best performing combination layer parameters (GCC and the Boolean NFL or GCL) had in every case a higher sensitivity and larger AUC value for the Cumulative Z-score approach. ROC curves of GL based on Cumulative Z-scores criterion are shown in Supplemental Figure S1.

The results of glaucoma sub-groups (early vs. moderate-advanced glaucoma) for diagnostic performance based on total abnormal pixel count (criteria 1 and 2) are listed in Table 4 and plotted as ROC curves in Figure 6. When dividing GL into early GL and moderate-advanced GL groups, the ranks of the best performing criteria are very similar between the early GL group and the moderate-advanced GL group as a whole (Table 4). For the Early GL group, again, the best performing criterion, with the highest AUC and sensitivity, was the Boolean combination criterion NFL or GCL (AUC 92.6%). The next-best performing criteria in the early GL group were GCC (AUC 90.4%) and any two of three layers (NFL, GCL, IPL, AUC: 90.6%). In the moderate-advanced group, the two highest AUC were found for the same combined criteria: NFL or GCL and any two of the three layers (NFL, GCL or IPL), which both had an AUC of 97.8%. The sensitivity was also highest for the combined criteria NFL or GCL in both early GL (73.3%) and moderate-advanced GL groups (95.1%). It should also be noted that the pattern of results was the same when the subgroups were defined using a cutoff of MD < −6 dB instead of −4 dB.

The outcomes based on analysis of cumulative Z-scores (criterion 3) for the two severity sub-groups considered separately are listed in Supplemental Table S1, with corresponding ROC curves shown in Supplemental Figure S2. The two best AUCs for early GL group and late GL group were also

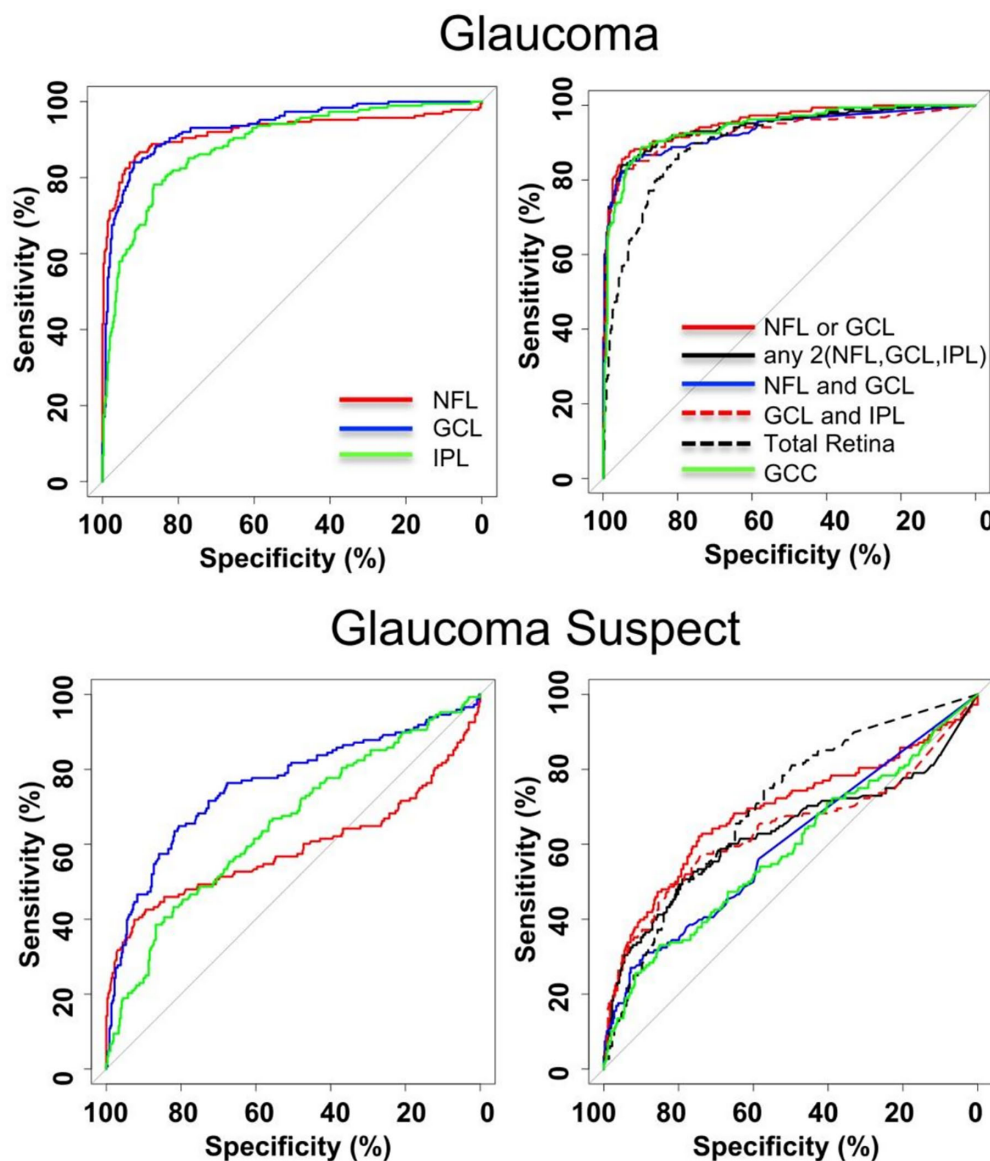


FIGURE 5. ROC curves for single layer thickness abnormalities or combinations of layer thickness abnormalities using abnormal pixel count criteria in glaucoma and glaucoma suspect eyes. Each curve is based on the total number of abnormal pixels counted in each eye for the designated layer or combination of layers. Values at each pixel for each thickness were adjusted to the mean age and Fovea-BMO distance of HC eyes. For each layer, a pixel with thickness values less than the fifth percentile of the HC distribution was defined abnormal.

found for the combined criteria NFL *or* GCL (early AUC 93.2%, late AUC 98.0%) and GCC (early AUC 90.9%, late AUC 98%). Although three combination criteria, NFL *or* GCL, any two of three layers (NFL, GCL, and IPL) and GCC were not significantly different from each other when compared across either of the GL severity subgroups or for the GL group overall.

Discriminating GLS From HC Eyes

The results for diagnostic performance based on total abnormal pixel count (Criteria 1 and 2) in GLS eyes are listed on the right half of [Figure 3](#) and plotted as ROC curves in the bottom panels of [Figure 5](#). In GLS eyes, the best performing criterion was GCL (AUC = 75.8%; Sensitivity 35.1%). The outcomes for GLS eyes based on analysis of cumulative Z-scores (criterion 3) are listed in Supplemental Table S1. The

pattern of results of the cumulative Z-score analysis of GLS eyes is similar insofar as GCL was best among all criteria, but not statistically better to performance using the total abnormal pixel count.

Continuous Sensitivity and AUC Maps for HC, GL, and GLS Eyes

To visualize the spatial “hot spots” of diagnostic performance and compare across parameters, we constructed continuous maps of AUC and sensitivity (at 95% specificity) values, which are shown in [Figure 7](#) and [Figure 8](#), respectively. These results reveal different spatial patterns of best diagnostic power depending on the individual inner retinal layer thickness being measured, with the strongest diagnostic information located where each layer is normally thickest in healthy eyes (NFL in arcuate pattern, GCL and IPL

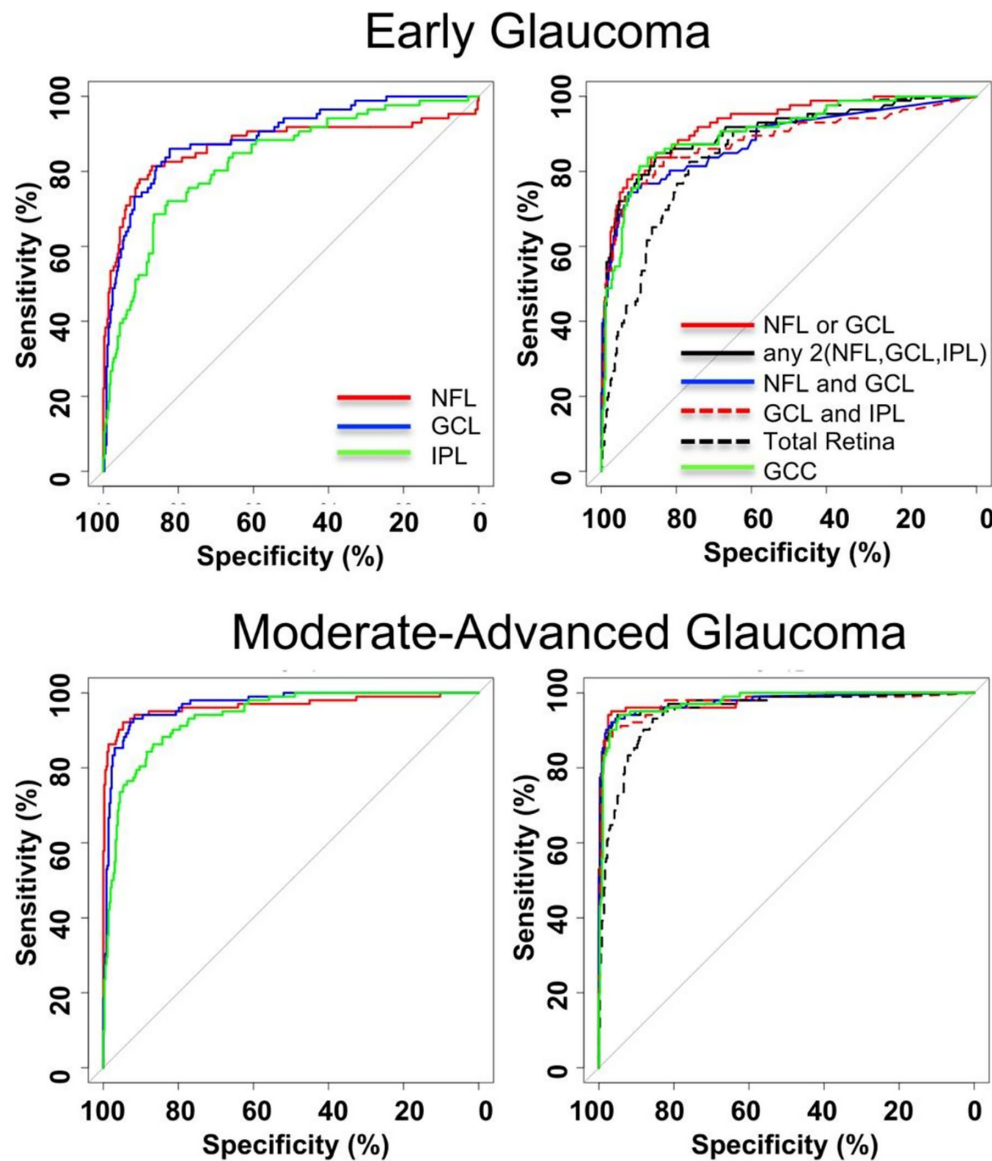


FIGURE 6. ROC curves for single-layer thickness abnormalities or combinations of layer thickness abnormalities using abnormal pixel count criteria in early glaucoma and moderate-advanced glaucoma eyes. Each curve is based on the total number of abnormal pixels counted in each eye for the designated layer or combination of layers. Values at each pixel for each thickness were adjusted to the mean age and Fovea-BMO distance of HC eyes. For each layer, a pixel with thickness values less than the fifth percentile of the HC distribution was defined as abnormal.

perifoveally). This spatial complementarity highlights one reason why a Boolean “or” criterion such as NFL or GCL, and a combined layer thickness parameter such as GCC tend to perform well. These results also show asymmetry across the horizontal midline with notably higher diagnostic sensitivity and performance for the inferior macula compared to the superior macula, for all three individual layers, NFL, GCL, and IPL.

DISCUSSION

The results of our study using pixel-based thickness measurements of individual inner retinal layers demonstrate that the diagnostic performance of the NFL and GCL exceeded that of the IPL but were similar to each other for detecting glaucomatous structural damage. Using thickness

measurements of combined layers and Boolean combinations of individual layers with the same pixel-based analyses, we found that the GCC has equivalent performance as NFL and GCL and that the Boolean combinations of individual layers provided only a modest improvement in diagnostic accuracy. As a result of using a pixel-based approach (rather than averaged thickness values from predetermined regions or sectors), we also found that the specific spatial locations within the posterior pole that exhibit best performance depend on which layer is being assessed. These results can be used by clinicians to select appropriate posterior pole OCT scan parameters and to refine their interpretation in the management of glaucoma. These results also suggest that the predetermined spatial patterns used commonly for sectoral thickness measurements are not ideally suited for all individual retinal layers or combined layer parameters.

TABLE 4. Single and Combined Layer Thickness Using Abnormal Pixel Count Parameter Detection Criteria in the Early GL and Moderate-Advanced GL Group

	Early GL MD ≥ −4 db (n = 84)		Moderate-Advanced GL MD < −4 db (n = 104)	
	Sensitivity at 95% Specificity	AUC	Sensitivity at 95% Specificity	AUC
Global single layer				
NFL	66.3 (51.2–77.9)	87.1 (81.5–92.7)	91.2 (83.3–96.1)	96.7 (94.2–99.2)
GCL	60.5 (45.3–74.4)	89.2 (85.2–93.2)	87.3 (79.4–94.1)	96.9 (95.3–98.5)
	(P = 0.5641)	(P = 0.4973)	(P = 0.3403)	(P = 0.8288)
IPL	39.5 (26.7–52.3)	82.8 (77.8–87.8)	73.5 (54.9–82.4)	93.6 (91.3–96)
	(P = 0.0072)	(P = 0.1647)	(P = 0.0293)	(P = 0.0224)
Combined layer				
GCC (all NFL, GCL, IPL)	59.3 (44.2–77.9)	90.4 (86.5–94.2)	93.1 (85.3–98)	97.7 (96.2–99.1)
	(P = 0.4762)	(P = 0.0479)	(P = 0.5226)	(P = 0.2099)
GCL and IPL	67.4 (54.6–80.2)	88.2 (83.3–93.2)	90.2 (84.3–96.1)	97.2 (95.1–99.3)
	(P = 0.9045)	(P = 0.7209)	(P = 1.0000)	(P = 0.6519)
NFL and GCL	68.6 (55.8–79.1)	88.2 (83.6–92.8)	93.1 (87.3–97.1)	97.7 (95.9–99.5)
	(P = 0.7128)	(P = 0.6404)	(P = 0.4907)	(P = 0.1434)
Any 2 of 3 (NFL, GCL, IPL)	70.9 (57–81.4)	90.6 (86.5–94.6)	93.1 (87.3–98)	97.8 (96.2–99.5)
	(P = 0.4716)	(P = 0.1972)	(P = 0.3716)	(P = 0.2150)
NFL or GCL	73.3 (60.5–83.7)	92.6 (89.5–95.7)	95.1 (90.2–99)	97.8 (96.1–99.5)
	(P = 0.3173)	(P = 0.0173)	(P = 0.2021)	(P = 0.1082)
TR	38.4 (23.3–50)	85 (80.8–89.2)	71.6 (58.8–84.3)	94.8 (92.4–97.1)
	(P = 0.0096)	(P = 0.4303)	(P = 0.0127)	(P = 0.1405)

AUC (%) and sensitivity (%) at 95% specificity as well as 95% confidence interval were listed in the table. Every criterion was compared against NFL and P value was listed.

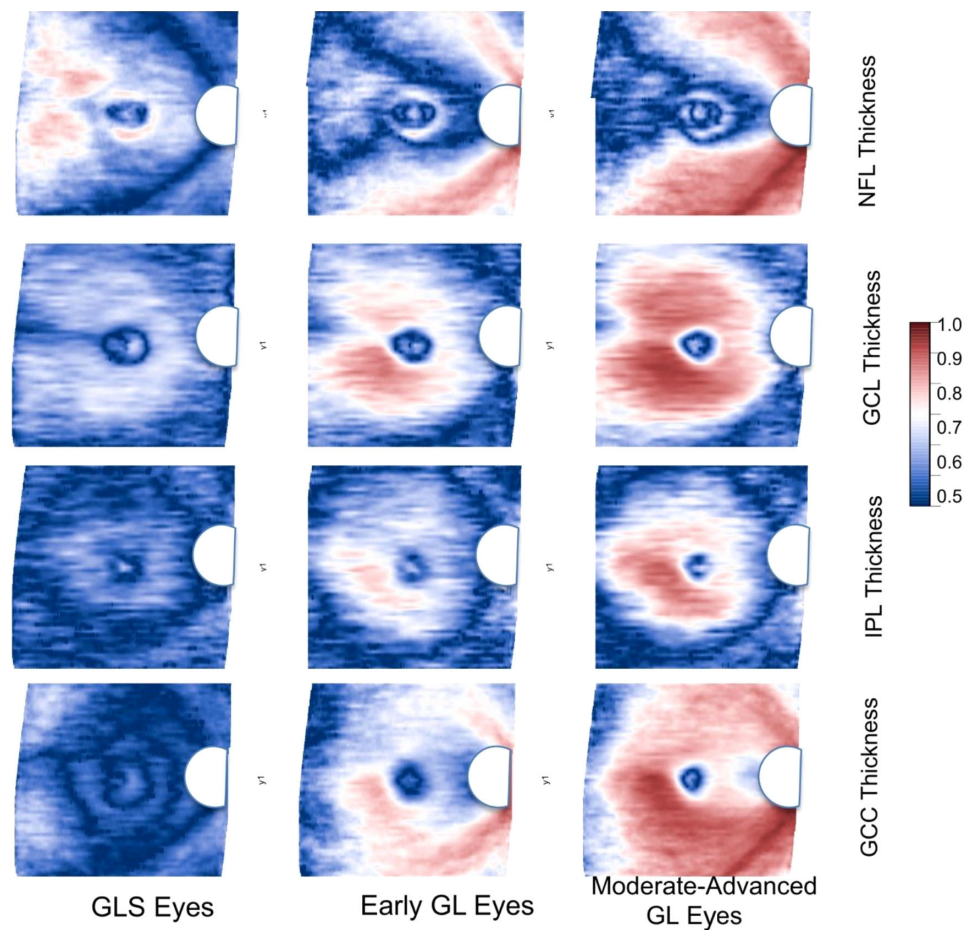


FIGURE 7. Continuous map of the AUC score of the individual layer thickness of glaucoma suspect, early glaucoma, and moderate-advanced glaucoma eyes. We used a boot-strapping method to calculate AUC value at each pixel location. The AUC value for each pixel was color-coded and printed on a continuous map with interpolated data from 768 × 61 pixels for each individual eye. We plotted all AUC values of all pixels within a standard rectangular area. For missing AUC values at the corners, the AUC values were not established because of missing values in healthy eyes (if ≥50% of the data was missing). From top row to bottom row: NFL thickness, GCL thickness, IPL thickness, and GCC layer.

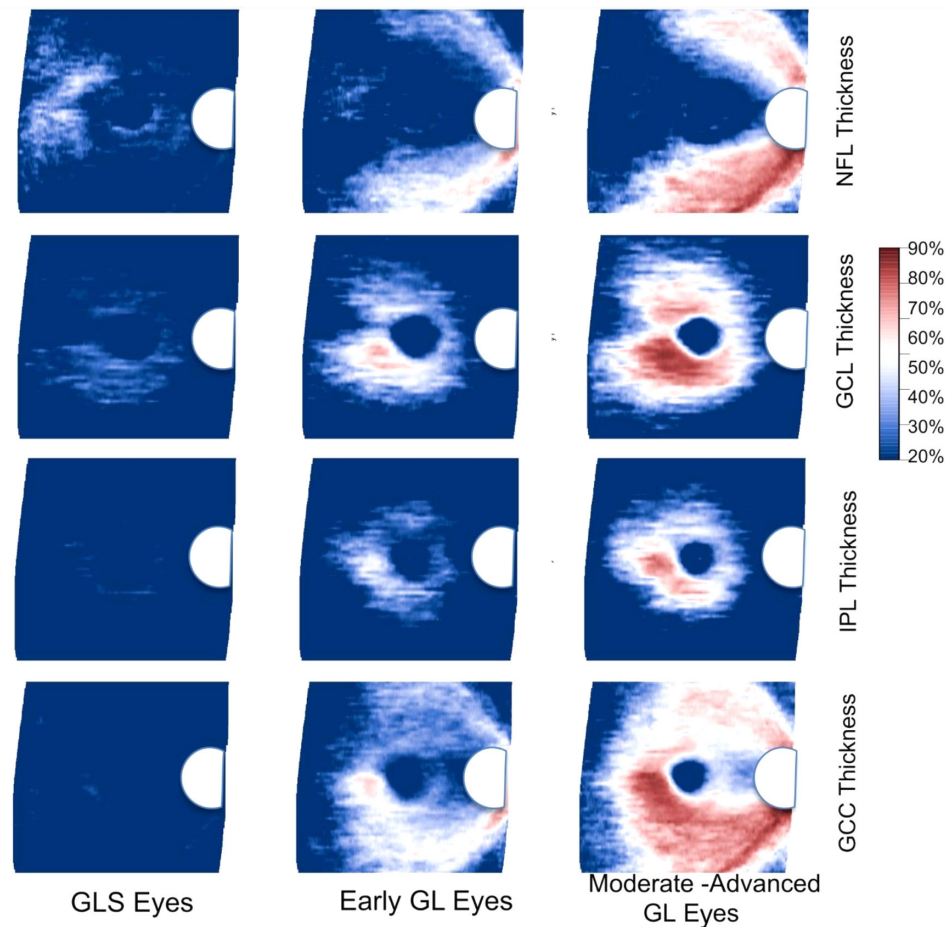


FIGURE 8. Continuous map of the sensitivity score at 95% specificity of the individual layer thickness of glaucoma suspect, early glaucoma, and moderate-advanced glaucoma eyes. We used a boot-strapping method to calculate sensitivity score at each pixel location. Sensitivity score for each pixel was color-coded and printed on a continuous map with interpolated data from 768×61 pixels for each individual eye. We plotted all sensitivity values of all pixels within a standard rectangular area. For missing sensitivity values at the corners, the sensitivity values were not established because of missing values in healthy eyes (if $\geq 50\%$ of the data was missing). From *top row to bottom row*: NFL thickness, GCL thickness, IPL thickness, and GCC layer.

The corollary is that either continuous, pixel-based measurements could be used (for any layer) or that the spatial sector pattern should be tailored to the desired layer.

Our first main finding implies that the thickness of the individual macular inner retinal layers NFL and GCL undergo thinning earlier and more frequently than the IPL as currently detected by the Spectralis OCT device we employed. However, while ganglion cell dendritic atrophy is known to occur at very early stages of experimental glaucoma and has even been shown to be a rescuable stage of retinal ganglion cell injury,²⁷ our results suggest that this pathophysiological phenomenon, to the extent that it may occur in clinical glaucoma,^{28,29} generally does not manifest as early thinning of the IPL, although it is possible that analysis of IPL sublaminae may reveal subtler changes.³⁰ Atrophy of retinal ganglion cell dendrites may also be masked by changes in other constituents of the IPL, such that textural analysis of OCT reflectance may reveal more than thickness measurements have done.³¹

Another finding of our study that is important to clinical diagnosis of glaucoma is that the best spatial locations for revealing glaucomatous loss within the macula depend on the specific layer being measured. The most salient locations for a given parameter do not necessarily conform to

the most common schemes of sectoral analysis in current use by commercial instrumentation. Although a “dartboard” like pattern of sectors centered on the fovea arrangement should work well for GCL (or GC-IPL) thickness, it would not be well suited for best detection of NFL thinning across the macula. Similarly, our results confirm a spatial pattern of earlier and more prevalent loss within the inferior macula compared to the superior macula,^{6,32} further supporting a benefit of comparing across the midline such as done in visual field analysis by the Glaucoma Hemifield Test and for asymmetry analysis of OCT posterior pole scans.^{32–34} Our results also lend further support for use continuous spatial maps and continuous probabilities (*z*-scores), much like the extensive work by Hood and colleagues has shown,^{35,36} rather than relying on color-coded broad sectors and/or discrete probability (or thickness) maps for glaucoma diagnosis based on OCT scans of the posterior pole.^{4,12,37–40} Our results showed slightly better performance of cumulative *z*-score compared to total abnormal pixel counts across all individual and combined layer definitions, which is consistent with our hypothesis that the *z*-score approach is more capable of detecting deepening localized loss. However, this improvement did not achieve statistical significance despite the relatively large size of our cohorts and was found only

for GL eyes, not GL suspect eyes. Thus the benefit of a z-score approach may be only marginal for population-based diagnosis and best suited to longitudinal follow-up within individual eyes (to better detect deepening of localized loss in addition to lateral spread of progressive damage).

Another result of our study demonstrated that there was a modest benefit to glaucoma diagnosis of analyzing individual inner retinal layer thickness across the posterior pole. Individual layer thickness was only a small enhancement over what is probably the most common parameter currently used, the GCC thickness, which combines the NFL, GCL and IPL together. Our finding that the IPL had substantially worse diagnostic performance than either NFL or GCL as individual single layers is the most likely explanation for why the GCC had slightly lesser performance than the Boolean combination NFL or GCL. However, this difference was small, not statistically significant and subject to the additional requirement of image segmentation for separating individual layers. More accurate segmentations may alleviate this burden in the future and enable larger databases to determine the potential diagnostic advantages with greater precision.

In our research, we are examining both individuals with diagnosed glaucoma and those who are considered potential glaucoma suspects. It is recognized that only a fraction of the eyes categorized as “glaucoma suspects”—between 30% and 60% are likely to develop POAG over the next 20 years and only about one fourth would be likely to develop detectable visual field damage over that time (given that the vast majority of these eyes had a similar profile to the OHTS inclusion criteria at study entry).⁴¹ Therefore a reliable diagnostic metric should demonstrate significantly improved performance in differentiating between “suspect” and “healthy” categories compared to random chance (e.g., an AUC well above 50%). However, this performance should be lower than the accuracy achieved in distinguishing between “glaucoma” and “healthy” eyes. Assessment of this group of eyes categorized as “suspect” provides an additional opportunity to compare the OCT parameters head-to-head for their relative diagnostic efficacy and as a test of the robustness of the findings from the eyes classified as having glaucoma. As seen in Table 3, all metrics were able to distinguish “glaucoma suspect” from “healthy” eyes, with AUC significantly greater than 50%. These AUCs were also in all cases lower than when distinguishing “glaucoma” from “healthy” eyes, and the pattern across OCT parameters was quite similar for the two study groups. These findings strengthen our conclusions, because the “glaucoma suspect” group likely contains a mix of truly glaucomatous and non-glaucomatous eyes, and the results suggest that the same OCT parameters may be helpful in detecting early structural damage and/or for stratifying risk in eyes categorized as being “glaucoma suspects.”

Among the limitations of this study, one important to consider is that only one eye of each participant was included in the analysis; so we did not perform intereye asymmetry analysis, for example, which may otherwise lead to different results. Further, it is likely that some of the GL suspect eyes have higher risk than others of having or developing glaucoma given a glaucoma diagnosis of their contralateral eye. While it is broadly true that such risk varies across any cohort of GL suspects based on a wide array of factors, the focus of this study was on comparison between diagnostic criteria rather than on comparison between GL and GL suspect eyes, or between these cohorts and those forming the basis for other reports. That is, the key results

are not the absolute values observed for AUC or sensitivity, but rather how those values compared across criteria. In the same sense, it remains to be determined whether our results will generalize to other cohorts, particularly those comprised of more diverse ethnic backgrounds.

Our research has other limitations. Because of the complex nature of health profiles in senior individuals, we have not adjusted for most risk factors or medication usage. These risk factors could include systemic hypertension, migraine, diet-controlled diabetes, peripheral vasospasm, and family history of glaucoma. Adjusting for medication use, including IOP-lowering and diabetic medications, was also not feasible due to the wide variety of medications and dosages commonly used by the elderly population.

An additional point of consideration about our study is the somewhat arbitrary cutoff used to populate the severity sub-groups into Early vs Moderate-Advanced GL. While this resulted in similar sample sizes for these two sub-groups, and confirmed the overall pattern of results found for the entire group, suggesting robustness, it is possible that other cutoffs may have revealed slightly different results. We defined early glaucoma in eyes with VF MD > -4.0 dB which represents less functional loss than is typically used to define a cutoff for “early” glaucoma (MD > -6.0 dB).^{42–44} Our aim of this subanalysis was to find the best performing parameters in an early stage of glaucoma; therefore we used a more stringent definition than -6 dB cutoff. Nevertheless, to be able to compare to other studies, we repeated the same severity sub-group analysis using the MD cutoff of -6 dB (instead of -4 dB) and found the same pattern of results entirely, suggesting they are robust.

One other potential limitation is that we did not separate the retinal vasculature (neither larger vessels nor microvasculature) in the thickness analyses. It is possible that such an approach would yield enhanced diagnostic performance, for example, in cases where layer segmentation boundaries (and thus thickness measurements) are strongly influenced by the presence of large vessels or given that layer boundaries based on OCT reflectance are strongly influenced by the retinal capillary plexuses. In particular context to our study, for example, the intermediate capillary plexus located at the posterior IPL boundary may exert a strong influence on boundary contrast, segmentation and thickness. We used the instrument’s posterior pole horizontal scan pattern for this study, which is a structural (reflectance) OCT scan only and thus does not provide OCT angiography results that would otherwise facilitate separate analysis in which vascular voxels are excluded. Future studies may help determine if such an approach would be diagnostically beneficial. Moreover, future studies may also determine the benefit of combining metrics from the posterior pole scan area with measurements of the peripapillary RNFL thickness, ONH rim, and lamina cribrosa, particularly using a topographically correspondent approach.^{45–47}

One difference in the approach we applied in this study from clinical practice is that we adjusted each thickness value at each pixel according to the linear functions relating thickness to age and to FoBMO distance in HC eyes, then determined diagnostic performance for each layer at each pixel using the adjusted values in all eyes. The adjustments for age were small for individual inner retinal layers (see Methods), but this differs from common clinical practice for which unadjusted values are compared against the commercial instrument’s internal normative range, itself either age-specific or age-adjusted. Although this may affect compar-

isons to other studies, the main point of our study was to compare diagnostic performance of layers and approaches to each other; thus it was important to apply the same approach across these conditions. Finally, we did not adjust transverse scaling according to axial length differences in this study. However, we excluded eyes with significant refractive error, so our results may not generalize to eyes with higher axial myopia. Moreover, we did adjust for the fovea-to-BMO centroid distance, itself a strong surrogate of the effects on lateral magnification of increased axial length.⁴⁸

In summary, in this study we compared the ability to discriminate GL and GLS eyes from healthy eyes on the basis of pixelwise OCT thickness measurements for three individual inner retinal layers, combined layers (such as GCC and total neural retina), as well as for Boolean combinations of individual layers. We used manually corrected segmentations of OCT scans in large groups of well-characterized HC eyes, GLS eyes and GL eyes. We found much better diagnostic performance for NFL and GCL than for IPL, suggesting that thinning of the NFL and GCL occurs earlier and more frequently than the IPL in glaucomatous neurodegeneration. However, diagnostic performance of NFL and GCL was only modestly improved (not statistically significant) over the commonly used combined GCC layer, suggesting that inclusion of the IPL in this parameter does not dilute its efficacy much. Our results demonstrate the importance of using continuous maps instead of the current common predetermined spatial patterns of sectors or grids. This is due to the fact that the most diagnostically useful pixels across the posterior pole do not necessarily conform to any of these patterns, depending on which layer is being assessed. Thus continuous maps of abnormality or cumulative Z-scores of individual layer thicknesses (NFL or GCL) or the combined layers (GCC parameter) may be best for clinical practice.

Acknowledgments

The authors thank Melinda Lin for her assistance with the manuscript final submission and Tony Yu for his help with revising tables. Heidelberg Engineering funded image acquisition at each of the eight study centers. Heidelberg Engineering provided additional unrestricted research support and technical assistance to Burgoyne and Chauhan, but played no role in data analysis, interpretation, presentation, or publication.

Supported by NIH/NIH/NEI R01-EY019674, R01-EY021281, R01-EY030590; Legacy Good Samaritan Foundation; Canadian Institutes of Health Research, Dalhousie Medical Research Foundation, Heidelberg Engineering, GmbH, Heidelberg.

Presented in part at the Association for Research in Vision and Ophthalmology Annual Meetings, May 2020, Virtual meeting.

Disclosure: **H. Yang**, Heidelberg Engineering (R); **J. Reynaud**, None; **G.P. Sharpe**, None; **D. Jennings**, None; **C. Albert**, None; **T. Holthausen**, None; **X. Jiang**, None; **S. Demirel**, None; **S.L. Mansberger**, Nicox (C), Allergan Research (C), Ocular Therapeutics (C); **M.T. Nicolela**, Heidelberg Engineering GmbH (R); Allergan (AbbVie) (R,C); Bausch + Lomb Canada (C); **S.K. Gardiner**, ONL Therapeutics (C); **B.C. Chauhan**, Revenio/CenterVue (C), Heidelberg Engineering (C), CenterVue (R), Heidelberg Engineering (R), Topcon (R); **C.F. Burgoyne**, Heidelberg Engineering (C, R); **B. Fortune**, Heidelberg Engineering (R), Perfuse Therapeutics, Inc. (C, F), Stoke Therapeutics, Inc. (F), Therapeutics, Inc. (C)

References

- Geevarghese A, Wollstein G, Ishikawa H, Schuman JS. Optical coherence tomography and glaucoma. *Annu Rev Vis Sci.* 2021;7:693–726.
- Tatham AJ, Medeiros FA. Detecting structural progression in glaucoma with optical coherence tomography. *Ophthalmology.* 2017;124:S57–S65.
- Rao HL, Pradhan ZS, Suh MH, Moghimi S, Mansouri K, Weinreb RN. Optical coherence tomography angiography in glaucoma. *J Glaucoma.* 2020;29:312–321.
- Pazos M, Dyrda AA, Biarnes M, et al. Diagnostic accuracy of Spectralis SD OCT automated macular layers segmentation to discriminate normal from early glaucomatous eyes. *Ophthalmology.* 2017;124:1218–1228.
- Kim KE, Park KH. Macular imaging by optical coherence tomography in the diagnosis and management of glaucoma. *Br J Ophthalmol.* 2018;102:718–724.
- Hood DC, Raza AS, de Moraes CG, Liebmann JM, Ritch R. Glaucomatous damage of the macula. *Prog Retin Eye Res.* 2013;32:1–21.
- Cifuentes-Canorea P, Ruiz-Medrano J, Gutierrez-Bonet R, et al. Analysis of inner and outer retinal layers using spectral domain optical coherence tomography automated segmentation software in ocular hypertensive and glaucoma patients. *PLoS One.* 2018;13:e0196112.
- Chua J, Tan B, Ke M, et al. Diagnostic ability of individual macular layers by spectral-domain OCT in different stages of glaucoma. *Ophthalmol Glaucoma.* 2020;3:314–326.
- Mwanza JC, Durbin MK, Budenz DL, et al. Glaucoma diagnostic accuracy of ganglion cell-inner plexiform layer thickness: comparison with nerve fiber layer and optic nerve head. *Ophthalmology.* 2012;119:1151–1158.
- Kim YK, Yoo BW, Jeoung JW, Kim HC, Kim HJ, Park KH. Glaucoma-diagnostic ability of ganglion cell-inner plexiform layer thickness difference across temporal raphe in highly myopic eyes. *Invest Ophthalmol Vis Sci.* 2016;57:5856–5863.
- Mohammadzadeh V, Fatehi N, Yarmohammadi A, et al. Macular imaging with optical coherence tomography in glaucoma. *Surv Ophthalmol.* 2020;65:597–638.
- Kim HJ, Lee SY, Park KH, Kim DM, Jeoung JW. Glaucoma diagnostic ability of layer-by-layer segmented ganglion cell complex by spectral-domain optical coherence tomography. *Invest Ophthalmol Vis Sci.* 2016;57:4799–4805.
- Wu Z, Weng DSD, Thenappan A, Ritch R, Hood DC. Evaluation of a region-of-interest approach for detecting progressive glaucomatous macular damage on optical coherence tomography. *Transl Vis Sci Technol.* 2018;7:14.
- Luo H, Yang H, Gardiner SK, et al. Factors influencing central lamina cribrosa depth: a multicenter study. *Invest Ophthalmol Vis Sci.* 2018;59:2357–2370.
- Yang H, Luo H, Gardiner SK, et al. Factors influencing optical coherence tomography peripapillary choroidal thickness: a multicenter study. *Invest Ophthalmol Vis Sci.* 2019;60:795–806.
- Gardiner SK, Boey PY, Yang H, Fortune B, Burgoyne CF, Demirel S. Structural measurements for monitoring change in glaucoma: comparing retinal nerve fiber layer thickness with minimum rim width and area. *Invest Ophthalmol Vis Sci.* 2015;56:6886–6891.
- Yang H, Luo H, Hardin C, et al. Optical coherence tomography structural abnormality detection in glaucoma using topographically correspondent rim and retinal nerve fiber layer criteria. *Am J Ophthalmol.* 2020;213:203–216.
- Gardiner SK, Johnson CA, Demirel S. Factors predicting the rate of functional progression in early and suspected glaucoma. *Invest Ophthalmol Vis Sci.* 2012;53:3598–3604.

19. Cohen JF, Korevaar DA, Altman DG, et al. STARD 2015 guidelines for reporting diagnostic accuracy studies: explanation and elaboration. *BMJ Open*. 2016;6:e012799.
20. Ren R, Yang H, Gardiner SK, et al. Anterior lamina cribrosa surface depth, age, and visual field sensitivity in the Portland Progression Project. *Invest Ophthalmol Vis Sci*. 2014;55:1531–1539.
21. Chauhan BC, Vianna JR, Sharpe GP, et al. Differential effects of aging in the macular retinal layers, neuroretinal rim, and peripapillary retinal nerve fiber layer. *Ophthalmology*. 2020;127:177–185.
22. Fortune B, Choe TE, Reynaud J, et al. Deformation of the rodent optic nerve head and peripapillary structures during acute intraocular pressure elevation. *Invest Ophthalmol Vis Sci*. 2011;52:6651–6661.
23. Hood DC, Fortune B, Mavrommatis MA, et al. Details of glaucomatous damage are better seen on OCT en face images than on OCT retinal nerve fiber layer thickness maps. *Invest Ophthalmol Vis Sci*. 2015;56:6208–6216.
24. Jonas JB, Yan YN, Zhang Q, et al. Retinal nerve fibre layer thickness in association with gamma zone width and disc-fovea distance. *Acta Ophthalmol*. 2022;100:632–639.
25. DeLong ER, DeLong DM, Clarke-Pearson DL. Comparing the areas under two or more correlated receiver operating characteristic curves: a nonparametric approach. *Biometrics*. 1988;44:837–845.
26. Pepe M, Longton G, Janes H. Estimation and comparison of receiver operating characteristic curves. *Stata J*. 2009;9:1.
27. Belforte N, Agostinone J, Alarcon-Martinez L, et al. AMPK hyperactivation promotes dendrite retraction, synaptic loss, and neuronal dysfunction in glaucoma. *Mol Neurodegener*. 2021;16:43.
28. Tribble JR, Vasalauskaite A, Redmond T, et al. Midget retinal ganglion cell dendritic and mitochondrial degeneration is an early feature of human glaucoma. *Brain Commun*. 2019;1:fcz035.
29. Mohammadi M, Su E, Chew L, et al. Comparison of ganglion cell layer and inner plexiform layer rates of change in suspected and established glaucoma. *Am J Ophthalmol*. 2023;249:12–20.
30. Ghassabi Z, Kuranov RV, Schuman JS, et al. In vivo sublayer analysis of human retinal inner plexiform layer obtained by visible-light optical coherence tomography. *Invest Ophthalmol Vis Sci*. 2022;63:18.
31. Morgan JE, Tribble J, Fergusson J, White N, Erchova I. The optical detection of retinal ganglion cell damage. *Eye (Lond)*. 2017;31:199–205.
32. Sullivan-Mee M, Ruegg CC, Pensyl D, Halverson K, Qualls C. Diagnostic precision of retinal nerve fiber layer and macular thickness asymmetry parameters for identifying early primary open-angle glaucoma. *Am J Ophthalmol*. 2013;156:567–577.e561.
33. Seo JH, Kim TW, Weinreb RN, Park KH, Kim SH, Kim DM. Detection of localized retinal nerve fiber layer defects with posterior pole asymmetry analysis of spectral domain optical coherence tomography. *Invest Ophthalmol Vis Sci*. 2012;53:4347–4353.
34. Takemoto D, Higashide T, Ohkubo S, Udagawa S, Sugiyama K. Ability of macular inner retinal layer thickness asymmetry evaluated by optical coherence tomography to detect preperimetric glaucoma. *Transl Vis Sci Technol*. 2020;9:8.
35. Hood DC, Slobodnick A, Raza AS, de Moraes CG, Teng CC, Ritch R. Early glaucoma involves both deep local, and shallow widespread, retinal nerve fiber damage of the macular region. *Invest Ophthalmol Vis Sci*. 2014;55:632–649.
36. Hood DC. Improving our understanding, and detection, of glaucomatous damage: an approach based upon optical coherence tomography (OCT). *Prog Retin Eye Res*. 2017;57:46–75.
37. Garcia-Medina JJ, Del-Rio-Vellosillo M, Palazon-Cabanes A, Pinazo-Duran MD, Zanon-Moreno V, Villegas-Perez MP. Glaucomatous maculopathy: thickness differences on inner and outer macular layers between ocular hypertension and early primary open-angle glaucoma using 8 × 8 posterior pole algorithm of SD-OCT. *J Clin Med*. 2020;9:1503.
38. Chien JL, Ghassabi MP, Patthanathamrongkasem T, et al. Glaucoma diagnostic capability of global and regional measurements of isolated ganglion cell layer and inner plexiform layer. *J Glaucoma*. 2017;26:208–215.
39. Kim HJ, Park KH, Kim YK, Jeoung JW. Evaluation of layer-by-layer segmented ganglion cell complex thickness for detecting early glaucoma according to different macular grids. *J Glaucoma*. 2017;26:712–717.
40. Mahmoudinezhad G, Mohammadzadeh V, Martinyan J, et al. Comparison of ganglion cell layer and ganglion cell/inner plexiform layer measures for detection of early glaucoma. *Ophthalmol Glaucoma*. 2023;6:58–67.
41. Kass MA, Heuer DK, Higginbotham EJ, et al. Assessment of cumulative incidence and severity of primary open-angle glaucoma among participants in the ocular hypertension treatment study after 20 years of follow-up. *JAMA Ophthalmol*. 2021;139:1–9.
42. Leung CKS, Guo PY, Lam AKN. Retinal nerve fiber layer optical texture analysis: involvement of the papillomacular bundle and papillofoveal bundle in early glaucoma. *Ophthalmology*. 2022;129:1043–1055.
43. WuDunn D, Takusagawa HL, Rosdahl JA, et al. Central visual field testing in early glaucoma: a report by the American Academy of Ophthalmology. *Ophthalmology*. 2024;131:240–248.
44. Dhabarde KA, Kende RP, Rahul NV, Surabhi, Nangare AR. Structure-function relationship and diagnostic value of macular ganglion cell complex measurement using Fourier-domain OCT in glaucoma. *Indian J Ophthalmol*. 2024;72:363–369.
45. Leung CK. Diagnosing glaucoma progression with optical coherence tomography. *Curr Opin Ophthalmol*. 2014;25:104–111.
46. Zheng F, Yu M, Leung CK. Diagnostic criteria for detection of retinal nerve fibre layer thickness and neuroretinal rim width abnormalities in glaucoma. *Br J Ophthalmol*. 2020;104:270–275.
47. Gmeiner JM, Schrems WA, Mardin CY, Laemmer R, Kruse FE, Schrems-Hoesl LM. Comparison of Bruch's membrane opening minimum rim width and peripapillary retinal nerve fiber layer thickness in early glaucoma assessment. *Invest Ophthalmol Vis Sci*. 2016;57:OCT575–OCT584.
48. Qiu K, Wang G, Zhang R, Lu X, Zhang M, Jansonius NM. Influence of optic disc-fovea distance on macular thickness measurements with OCT in healthy myopic eyes. *Sci Rep*. 2018;8:5233.